

Gautham Vasan

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Research Interests

I'm broadly interested in building agents that can continually learn from experience, adapt, and improve over their lifetimes. To this end, I have actively worked on online reinforcement learning, imitation learning, and reward design for sequential decision-making problems, with applications in robotics.

Education

- Ph.D. in Computing Science**, University of Alberta, Canada 2020 - Present
- Supervisory Committee: Dr. Rupam Mahmood (advisor), Dr. Richard Sutton, Dr. Matthew Taylor
 - Dissertation (proposed): Real-Time Reinforcement Learning For Robots
- M.Sc. in Computing Science (Thesis)**, University of Alberta, Canada 2017
- Advisor: Dr. Patrick Pilarski
 - Dissertation: [Teaching a Powered Prosthetic Arm with an Intact Arm Using Reinforcement Learning](#)
 - Winner of the M.Sc Outstanding Thesis Award in Computing Science
- B.Tech. in Instrumentation and Control Engineering**, National Institute of Technology, Tiruchirappalli, India 2015
- Advisors: Dr. G. Saravana Ilango, Dr. V. Sankaranarayanan
 - Capstone Project: [Autonomous Visual Tracking and Landing of a Quadrotor on a Moving Platform](#)

Employment

- Zyphra Technologies**, Member of Technical Staff, San Francisco, USA 02/2026 - Present
- Developing new model architectures for long-context reasoning, long-term memory, and continual learning.
 - Designing computationally efficient learning algorithms for training large language models (LLMs) and multi-modal world models.
- Sanctuary AI**, Research Scientist Intern, Vancouver, Canada 05/2025 - 09/2025
- Design and development of novel reinforcement learning algorithms that can leverage expert demonstrations for real-world manipulation tasks.
 - Benchmarking of *vision-language-action models*, *diffusion policies*, and *imitation-bootstrapped RL* for diverse robotic platforms, with applications to Sanctuary's humanoid Phoenix.
- University of Freiburg**, Visiting Researcher/DAAD Scholar, Freiburg im Breisgau, Germany 03/2023 - 06/2023
- Hosted by Dr. Joschka Boedecker as a part of a DAAD-Stiftung Unicore Fellowship.
 - Reinforcement learning from human feedback for integrating noisy electroencephalogram (EEG) signals decoded from a patient's brain into a framework for skill learning on assistive robots.
- Ocado Technology (Formerly Kindred AI)**, Machine Learning Researcher, Toronto, Canada 2017 - 2020
- Devised deep reinforcement learning techniques for [SORT](#), a [piece-picking robot](#) that grasps, scans and stows items in warehouses for clothing stores like GAP and American Eagle.
 - Developed [SenseAct](#), an open-source computational framework for reliable, reproducible learning on physical robots.
 - Deployed [RLScan](#) to production. RLScan uses deep RL to learn a vision-based policy to scan barcodes on clothing with a Fanuc arm. It was trained end-to-end in production, learning from a fleet of robots across multiple warehouses.

Publications

Peer-Reviewed Publications

17. Fahim Shahriar, Cheryl Wang, Alireza Azimi, [Gautham Vasan](#), Hany Elanwar, Abhishek Naik, A. Rupam Mahmood, Colin Bellinger, [Dynamic Object Masks as Goal Representations for Visual Goal-Conditioned Reinforcement Learning](#). *RLC 2026*.
16. [Gautham Vasan](#), Mohamed Elsayed, Alireza Azimi, Jiamin He, Fahim Shahriar, Colin Bellinger, Martha White, A. Rupam Mahmood, [Deep Policy Gradient Methods Without Batch Updates, Target Networks, or Replay Buffers](#). *NeurIPS 2024 (Video/Code)*
15. Mohamed Elsayed, [Gautham Vasan](#), A. Rupam Mahmood, [Streaming Deep Reinforcement Learning Finally Works](#). *NeurIPS FITML Workshop 2024*, ([Code](#))

14. [Gautham Vasan](#), Yan Wang, Fahim Shahriar, James S. Bergstra, Martin Jagersand, A. Rupam Mahmood, [Revisiting Sparse Rewards for Goal-Reaching Reinforcement Learning](#). *RLC 2024 (Video/Code)*
13. Huiyi Wang, Fahim Shahriar, Alireza Azimi, [Gautham Vasan](#), A. Rupam Mahmood, Colin Bellinger, [Versatile and Generalizable Manipulation via Goal-Conditioned Reinforcement Learning with Grounded Object Detection](#). *CoRL MRM-D Workshop 2024*
12. [Gautham Vasan](#). [Autonomous Skill Acquisition for Robots Using Graduated Learning](#). *AAMAS 2024*
11. Bram Grooten, Tristan Tomilin, [Gautham Vasan](#), Matthew E. Taylor, Rupam Mahmood, Meng Fang, Decibal Mocanu, [MaDi: Learning to Mask Distractions for Generalization in Visual Deep Reinforcement Learning](#). *AAMAS 2024 (Video/Code)*
10. [Gautham Vasan*](#), Yan Wang*, Fahim Shahriar, James S. Bergstra, A. Rupam Mahmood, [Learning Sparse Reward Tasks on Real Robots From Scratch](#), *RAP4 Robotics Workshop, ICRA 2023*
9. Fengdi Che, [Gautham Vasan](#), A. Rupam Mahmood, [Correcting discount-factor mismatch in on-policy policy gradient methods](#), *ICML 2023*
8. Yan Wang*, [Gautham Vasan*](#), A. Rupam Mahmood, [Real-Time Reinforcement Learning for Vision-Based Robotics Utilizing Local and Remote Computers](#), *ICRA 2023 (Video/Code)*
7. Dmytro Korenkevych, A. Rupam Mahmood, [Gautham Vasan](#), James Bergstra, [Autoregressive policies for continuous control deep reinforcement learning](#), *IJCAI 2019 (Video/Website)*
6. A. Rupam Mahmood, Dmytro Korenkevych, [Gautham Vasan](#), William Ma, James Bergstra, [Benchmarking reinforcement learning algorithms on real-world robots](#), *CoRL 2018 (Video/Code/Website)*
5. [Gautham Vasan](#), Patrick M. Pilarski, [Context-Aware Learning from Demonstration: Using Camera Data to Support the Synergistic Control of a Multi-Joint Prosthetic Arm](#), *IEEE BioRob 2018*
4. [Gautham Vasan](#), Patrick M. Pilarski, [Learning from Demonstration: Teaching a Myoelectric Prosthesis with an Intact Limb via Reinforcement Learning](#), *IEEE ICORR 2017 (Video)*
Top 10% of submissions. Selected for oral presentation.
3. Kenny Young, [Gautham Vasan](#), Ryan Hayward, [NeuroHex: A Deep Q-learning Hex Agent](#), *Computer Games Workshop at IJCAI 2016*
2. Juhi Ajmera, Siddharthan P. R., Ramaravind K. M., [Gautham Vasan](#), Naresh Balaji R. and V. Sankaranarayanan, [Autonomous visual tracking and landing of a quadrotor on a moving platform](#), *IEEE ICIIP 2015 (Video)*
1. [Gautham Vasan](#), Naresh Balaji Ravichandran, Gowtham Kumar T.S.B, Aravind Govindan, G Saravana Ilango, [A Control Strategy for an Autonomous Robotic Vacuum Cleaner for Solar Panels](#), *Texas Instruments Educators Conference 2014 (Video)*

Pre-Prints

- Joydeep Biswas, Sheila Schoepp, [Gautham Vasan](#), Anthony Pipari, Arthur Zhang, Zichao Hu, Sebastian Joseph, Matthew Lease, Junyi Jessy Li, Peter Stone, Kiri L. Wagstaff, Matthew E. Taylor, Odest Chadwicke Jenkins. [AI-Assisted Peer Review at Scale: The AAIL-26 AI Review Pilot](#). arXiv preprint arXiv:2604.13940, 2026
- Qingeng Lan, [Gautham Vasan](#), A. Rupam Mahmood, [Efficient Reinforcement Learning by Reducing Forgetting with Elephant Activation Functions](#). arXiv preprint arXiv:2509.19159, 2025.

Peer-reviewed Abstracts

- [Gautham Vasan](#), Patrick M. Pilarski, [Mirrored Bilateral Training of a Myoelectric Prosthesis with a Non-Amputated Arm via Actor-Critic Reinforcement Learning](#), *Reinforcement Learning and Decision Making (RLDM) 2017*.
Top 8% of submissions. Selected for oral presentation.
- Craig Sherstan, Marlos C. Machado, Jaden Travník, Adam White, [Gautham Vasan](#), Patrick M. Pilarski, [Confident Decision Making with General Value Functions](#), *Reinforcement Learning and Decision Making (RLDM) 2017*.

Thesis

- [Gautham Vasan](#), Examining Committee: Patrick M. Pilarski, Martha White and K Ming Chan, [Teaching a Powered Prosthetic Arm with an Intact Arm Using Reinforcement Learning](#), M.Sc Thesis, University of Alberta, Edmonton, Canada, 2017.

Awards & Achievements

- AAMAS Scholarship to present research at the Doctoral Consortium (2024)
- DAAD-Stiftung UNICORE Scholarship for a three-month research visit to the University of Freiburg (2023)
- DAAD AINet Postdoctoral Networking Fellowship (2022) to visit and foster collaborations with German research labs
- University of Alberta Doctoral Recruitment Scholarship Fall 2020/21
- Winner of the M.Sc Outstanding Thesis Award in Computing Science at the University of Alberta (2017)
- Phase-1 Winners and Finalist at the Texas Instruments Innovation Challenge India Design Contest (2014)

Technical Skills

- Programming: Python, C++
- Tools: PyTorch, Jax, ROS, Docker
- Simulators: MuJoCo, Nvidia Isaac Sim, Gazebo
- Research Areas: Reinforcement Learning, Deep Learning, Robotics, Generative World Models, Real-Time Systems
- Robot Platforms: Franka Emika Panda, UR5, iRobot Create2, Anki Vector, Peto Bittle Quadruped, Dynamixel Servos

Conference & Workshop Organization

- Workflow Chair, AAAI 2026
 - Played a key role in launching AAAI's first AI-powered peer-review assessment system, a large-scale deployment of LLMs to support scientific review with human-in-the-loop safeguards

Professional Activities

- Reviewer: ICRA 2026 | ICLR 2026 | ICML 2025 | RLC 2025 | Collas 2025 | ICLR 2025 | NeurIPS 2023, 2024 | IEEE BioRob 2024, 2018 | IEEE ICDL 2024 | IROS 2023, 2020
- Candidate selection for the CIFAR Deep Learning and Reinforcement Learning Summer School 2023 & 2024
- Mentoring: Seven students at the University of Alberta (undergraduate and masters level) on robot learning research

Selected Talks

- *Streaming Deep Reinforcement Learning*, Cohere For AI, 28 Jan 2025 (Invited)
- *Deep Policy Gradient Methods Without Batch Updates, Target Networks, or Replay Buffers*, ML Collective, 7 Feb 2025. Also presented earlier at Mila, McGill University, Brown University and IIT Madras (Invited)
- *From Q-learning to Dreamer*, Amii Tea Time Talks, University of Alberta, 27 Aug 2024
- *Two Issues of Autonomous Robot Learning*, Amii AI Seminar, University of Alberta, 27 Oct 2023
- *Reward (Mis-)Specification in Reinforcement Learning*, Amii Tea Time Talks, 23 Aug 2023
- *Reinforcement Learning for Robots*, natChat @NeurAlbertaTech, 16 Feb 2023 (Invited)
- *Learning from Demonstration: Teaching a Myoelectric Prosthesis using an intact Limb via Reinforcement Learning*, Cognition Seminar, Dept. of Psychology, University of Alberta, 3 Feb 2017 (Invited)

Relevant Coursework

Graduate: Deep Policy Gradient Methods | Theoretical Foundations of Reinforcement Learning | Statistical Computing | Machine Learning and The Brain | Introduction to Reinforcement Learning | Introduction to Machine Learning | Convolutional Neural Nets for Image Processing | Actor-Critic Algorithms | Medical Robotics and Computer Assisted Surgery

Undergraduate: Linear Algebra and Probability Theory | Digital Signal Processing | Numerical Methods | Data Structures and Algorithms | Signals and Systems | Sensors and Transducers | Control Systems | Neural Networks and Fuzzy Logic

Teaching Experience

- CMPUT 340: Introduction to Numerical Methods (Winter 2024)
- CMPUT 653: Real-Time Policy Learning (Fall 2023)
- CMPUT 365: An Introduction to Reinforcement Learning (Winter 2021, Winter 2022, Fall 2022)
- CMPUT 174: Introduction to the Foundations of Computation I (Fall 2015, Winter 2016, Fall 2020)

Service & Outreach

- *Volunteer:* DiscoverE Summer Camp 2023, to showcase and explain robotics research and its real-world applications to Grade 4-6 students.
- Research Volunteer, The Hospital for Sick Children (SickKids, 2019)
- Conducted medical trials with patients with cerebral palsy and spasticity to assess the feasibility of controlling assistive robots using myoelectric signals (2017).

Personal

- **Citizenship:** Canada

References

Available upon request.